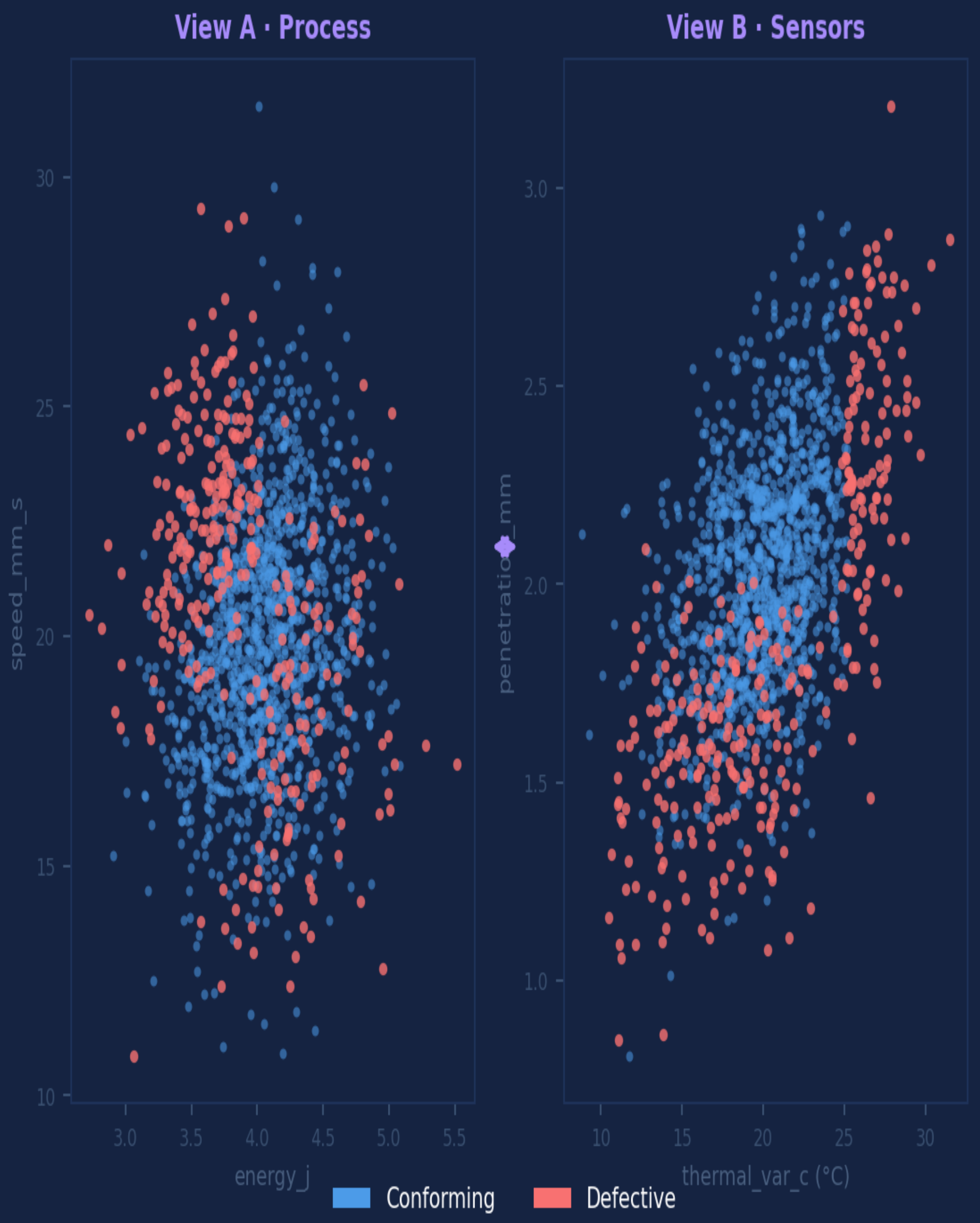
Laser Weld
Defect Detection

Co-Training · Semi-Supervised

Project 14 · LozanoLsa Industrial ML Portfolio

Two views. Two models.
One shared question — is this weld good?

Python | scikit-learn | LogReg | Co-Training



1,912
Laser Welds

150
Labeled (8%)

0.704
AUC-ROC Ensemble

77.8%
Recall — Defects

The Real Problem

Only 8% of welds are ever labeled

Manual inspection — destructive testing, CT, ultrasound cross-sections — covers at most 8% of production volume. The other 1,762 welds passed through the line unclassified. Their sensor data is complete. Their quality status is unknown.

Two independent data streams exist for every weld

View A (process controller) and View B (in-process sensors) run simultaneously, collected by independent instrument chains. This structural independence is not a coincidence — it is Co-Training's operating condition.

Supervised models discard 90% of what they know

A purely supervised approach trains on 120 examples and ignores 1,762. The acoustic signals, thermal traces, and penetration readings from those unlabeled welds contain defect signal — it just has no label attached yet.

150 labeled welds · 1,762 unlabeled · 10 features across 2 independent views

Full inspection of every weld is physically impossible at production line speed. The sensor data exists — the labels don't.

From Single-View Supervision to Cross-View Teaching

⊘ Old Thinking

— Label 8% of welds through destructive inspection.

— Train a single model on 120 examples.

— Discard 1,762 welds with complete sensor data.

— Accuracy is limited by label scarcity.

— Both views are available — only one is used.

— No mechanism to improve without more inspection.

✓ New Thinking

✓ Model A (process) teaches Model B (sensors).

✓ Model B teaches Model A — never itself.

✓ Confidence threshold 0.65 gates pseudo-labels.

✓ Round 1 alone adds 2,263 new training examples.

✓ Both views contribute. Neither dominates alone.

✓ Evaluate only on real labels. Always.

Co-Training does not teach itself. Each model only receives pseudo-labels donated by the other — that constraint is what makes it trustworthy.

The Data Behind the Problem — Two Views

View A — Process Parameters

Laser Controller

IPG / TRUMPF

energy_j · speed_mm_s · frequency_khz · spot_diameter_mm

Set by the CNC program — the intended process conditions.

Gas MFC

Flow Controller

gas_flow_l_min

Shielding gas metered by mass flow controller → SCADA.

View B — Sensor Response

Profilometer & Ultrasound

Keyence / Olympus

bead_height_mm · penetration_mm

Geometric weld quality — the physical outcome of the melt pool.

Pyrometer & Piezo

Raytek / DAQ

thermal_var_c · acoustic_rms · vibration_rms

Process stability signals — thermal and mechanical fingerprint of fusion.

Both views collect on every weld, every shift. Co-Training uses that parallel structure as a pedagogical channel.

Why Co-Training?

"The process view and the sensor view emerged from different physics. When they agree on a prediction, that agreement carries more information than either view alone."

01 True view independence

View A comes from the machine controller CAN bus. View B comes from independent sensor instruments — profilometer, pyrometer, piezo, ultrasound. Different physical principles. Different failure modes. Different signal paths.

02 Cross-teaching, not self-teaching

Model A only donates pseudo-labels to Model B's training set, and vice versa. Neither model teaches itself. This asymmetry prevents the circular error propagation that Self-Training manages only through confidence gating.

03 Calibrated base learner

Logistic Regression produces well-calibrated probability estimates — P=0.70 means 70% of those cases are actually defective. The confidence threshold (0.65) is meaningful precisely because the model doesn't overfit its own certainty.

04 Ensemble as final answer

The two final models are averaged at inference. Neither view dominates — they vote together. A weld that one view flags and the other misses lands at ~0.50 probability, the natural uncertainty zone. Consensus defects carry the highest operational confidence.

Co-Training bets that two independent views will agree more when they are right than when they are wrong. On this welding line, the bet held.

Model Results

73.3%
Accuracy — Ensemble

0.704
AUC-ROC — Ensemble

77.8%
Recall — Ensemble

View A · Process Drivers

speed_mm_s	+0.36
energy_j	−0.17
frequency_khz	−0.09
spot_diameter_mm	−0.01
gas_flow_l_min	−0.13

View B · Sensor Drivers

penetration_mm	+0.34
bead_height_mm	−0.13
thermal_var_c	+0.16
vibration_rms	+0.34
acoustic_rms	−0.10

Ensemble evaluated on 30 held-out real-labeled welds. Pseudo-labels never entered the test set. Recall 77.8% — 7 of 9 defective welds caught.

From Insight to Action

View A · speed_mm_s +0.36

Speed Cap SPC Rule

→ Add an SPC control limit at nominal +5% travel speed. High-speed passes are the #1 defect predictor. A speed exceedance alarm triggers process hold before the bead solidifies.

View B · penetration_mm +0.34

Penetration Floor Limit

→ Inline ultrasound reject threshold: any weld with measured penetration below 80% of nominal triggers automatic re-weld cycle. The deepest protective signal in View B.

View A · energy_j −0.17

Energy-Speed Interlocked Recipe

→ Parametric interlock: when speed increases above nominal, laser energy scales proportionally to maintain energy density per mm. Prevents the destructive speed-without-energy combination.

View B · thermal_var_c +0.16

Pyrometer Variance Monitoring

→ Real-time thermal variance window: if pyrometer variance exceeds threshold for 3 consecutive pulses, flag for immediate visual inspection. Unstable melt pool — catch it before it cools.

View B · vibration_rms +0.34

Fixture Vibration Pre-check

→ Add vibration RMS baseline check at fixture clamp-up. Elevated vibration before the weld begins predicts solidification disruption. Correct the fixture before igniting the laser.

Co-Training reveals which process lever — speed, energy, or both — needs correction before the defect is welded in.

What-If Simulator

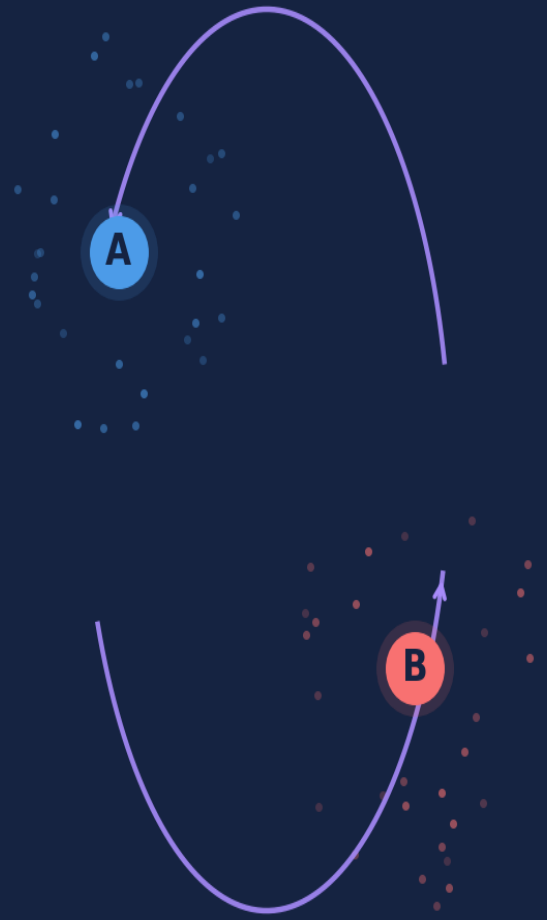
Scenario A · High-Risk Weld	
View A — Process	
Energy	**3.2 J**
Speed	**25.0 mm/s**
Frequency	**7.5 kHz**
Spot Ø	**0.75 mm**
Gas	**11.0 L/min**
View B — Sensors	
Bead H.	**1.05 mm**
Thermal	**28.0 °C**
Acoustic	**0.62**
Penetr.	**1.65 mm**
Vibration	**0.32**
P(defect) = 93%	

Scenario B · Conforming Weld	
View A — Process	
Energy	**4.4 J**
Speed	**17.0 mm/s**
Frequency	**8.0 kHz**
Spot Ø	**0.80 mm**
Gas	**12.0 L/min**
View B — Sensors	
Bead H.	**1.22 mm**
Thermal	**20.0 °C**
Acoustic	**0.60**
Penetr.	**2.4 mm**
Vibration	**0.30**
P(defect) = 32%	

Scenario C · Energy Corrected	
View A — Process	
Energy	**4.3 J ✓**
Speed	**25.0 mm/s**
Frequency	**7.5 kHz**
Spot Ø	**0.75 mm**
Gas	**11.0 L/min**
View B — Sensors	
Bead H.	**1.12 mm**
Thermal	**24.0 °C**
Acoustic	**0.61**
Penetr.	**1.85 mm**
Vibration	**0.31**
P(defect) = 77%	

Scenario C: restoring energy from 3.2 J to 4.3 J reduces ensemble risk from 93% to 77% — but thermal instability keeps it elevated.

Neither model was smarter. They were just looking at different things.



Co-Training rests on one bet: that two genuinely independent descriptions of the same event will agree more often when they are right than when they are wrong.
On this welding line, that bet held. The process view and the sensor view taught each other 1,414 pseudo-labels across 5 rounds — from 120 real examples.

Ensemble AUC-ROC 0.704. Recall 77.8%. Starting from 8% labeled data.

Full dataset (1,912 welds) · app.py dual-view simulator · PPTX deck
Try this: raise the confidence threshold to 0.80. Does Model B still converge?
Because a weld only gets one chance to be right.